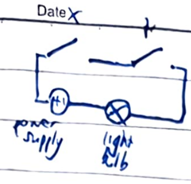
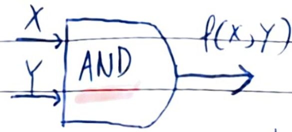
Digital Fundamentals - Logic Gates

Boolean Functions



x AND Y

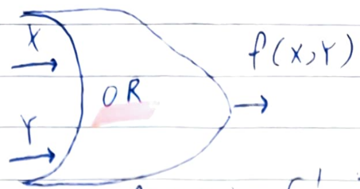
X	Y	f
0	0	0
0	1	0
1	0	0
1	1	1



$$f(X,Y) = \begin{cases} 1 & \text{when } x==1 \ \& \ y==1 \\ 0 & \text{otherwise} \end{cases}$$

X OR Y

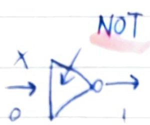
X	Y	OR
0	0	0
0	1	1
1	0	1
1	1	1

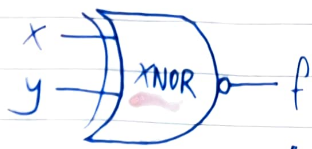


$$f(X,Y) = \begin{cases} 1 & \text{when } x==1 \text{ OR } y==1 \\ 0 & \text{otherwise} \end{cases}$$

Inverts

X	Not
0	1
1	0



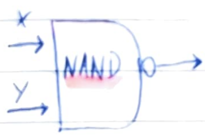


X	Y	f
0	0	1
0	1	0
1	0	0
1	1	1

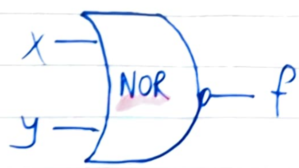
[2 1s or 0s]

Inverts AND Gate

X	Y	NAND
0	0	1
0	1	1
1	0	1
1	1	0



NOT + AND GATE



x	y	f
0	0	1
0	1	0
1	0	0
1	1	0

X Xor Y

x	y	Xor
0	0	0
0	1	1
1	0	1
1	1	0

1 if different inputs